Experiment 3
Pre-processor and Time Complexity

Aim / Objective:

The aim of studying **Pre-processor directives** and **Time Complexity** is to develop a deeper understanding of how programs are optimized and analyzed in terms of performance. Preprocessor directives in C, learning Time Complexity allows us to evaluate the efficiency of algorithms by measuring how the running time grows with input size, using notations like Big O. Together, these concepts equip students with essential tools for writing optimized, maintainable, and highperforming code.

Problem Statement:

1. **Define and use macros using #define and include header files with #include.**

   Show conditional compilation with #ifdef.
2. **Design C program to test library function math.h, ctype.h, string.h**

Solutions:

1. **Define and use macros using #define and include header files with #include.**

Algorithm:

Flowchart:

Show conditional compilation with #ifdef.

```
#include <stdio.h>
#define PI 3.14159
#define AREA(l,b) ((l)*(b))
#define DEBUG

#ifdef DEBUG
   #define LOG(x) printf("DEBUG: %s\n", x)
#else
   #define LOG(x)
#endif

int main() {     float len =
5.0, br = 2.0;     float area =
AREA(len,br);
   LOG("Calculating area of rectangle");     printf("Area
of rectangle: %.2f\n", area);     return
0;
}
```

Output:

```
Output

DEBUG: Calculating area of rectangle
Area of rectangle: 10.00


=== Code Execution Successful ===
```

2. Design C program to test library function math.h, ctype.h, string.h Solution :

Algorithm:

Flowchart:

```
#include <stdio.h>
#include <math.h>     // For mathematical functions
#include <ctype.h>    // For character functions
#include <string.h>   // For string functions

int main() {
   // ----- math.h -----     double num =
25.0;
printf("Using math.h:\n");
   printf("Square root of %.2f is %.2f\n", num, sqrt(num));
printf("Power: 2^3 = %.0f\n", pow(2, 3));
   printf("Sine of 0 radians = %.2f\n\n", sin(0));

   // ----- ctype.h -----     char ch1 = 'A', ch2 = '7', ch3 = ' ';     printf("Using ctype.h:\n");
printf("Is '%c' an alphabet? %s\n", ch1, isalpha(ch1) ?
"Yes" : "No");     printf("Is '%c' a digit? %s\n", ch2, isdigit(ch2) ? "Yes" :
"No");     printf("Is '%c' a whitespace? %s\n\n", ch3, isspace(ch3) ? "Yes" :
"No");

   // ----- string.h -----     char str1[] = "Hello";     char str2[] =
"World";     char str3[20];     printf("Using string.h:\n");
printf("Length of str1 = %lu\n", strlen(str1));     strcpy(str3,
str1);  // Copy str1 into str3     strcat(str3, str2);  // Concatenate
str2 to str3     printf("After copying and concatenating: %s\n",
```

```
str3);     printf("Comparing str1 and str2: %d\n", strcmp(str1,
str2));

   return 0;
}
```

Output:

```
Output
Using math.h:
Square root of 25.00 is 5.00
Power: 2^3 = 8
Sine of 0 radians = 0.00

Using ctype.h:
Is 'A' an alphabet? Yes
Is '7' a digit? Yes
Is ' ' a whitespace? Yes

Using string.h:
Length of str1 = 5
After copying and concatenating: HelloWorld
Comparing str1 and str2: -15


=== Code Execution Successful ===
```

Compilation and Execution Steps:

1. Open Dev C++.
2. Click File → New → Source File.
3. Type or paste your C code.
4. Click File → Save As and save the file with a .c extension.
5. Click Execute → Compile (or press **F9**).

6. If there are no errors, click Execute → Run (or press **Ctrl + F10**).
7. View output in the console window.

Practice Questions:

1. **Write a program to define a macro-SQUARE(x) and use it to calculate the square of 5.**
2. **Write a code snippet that prints "Debug Mode" only if DEBUG is defined using #ifdef.**

Learning Outcomes: Students will able to
LO1 Understand Pre-Processor and Time Complexity.

Course Outcome: After completion of course, the students will be able to understand Pre-processor and Time Complexity in C.

Conclusion:

In conclusion, **Pre-processor directives** and **Time Complexity** are two essential concepts in programming and algorithm analysis. Pre-processor directives are powerful tools in C that allow code modularity, macro definitions, conditional compilation, and more efficient management of large programs before actual compilation takes place. **Time Complexity** is crucial for evaluating and comparing the performance of algorithms.

Viva Questions:

1. What is the purpose of #include in a C program?
2. Explain the use of #ifdef, #ifndef, and #endif.

For Faculty Use only

Correction Parameters	Formative Assessment [40%]	Timely completion of Practical [40%]	Attendance / Learning Attitude [20%]	Total	Sign
Marks Obtained					